Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 2 Random variable and their probability distributions

Section 2.7 The independence of random variables

Independence is very important in probability and statistics. The concept of independence of two events in chapter 1 is to carry over the concept of independence from events to random variables.

The definition of two events ***A*** and ***B*** independent of each other, *if and only if*

$$P(\boldsymbol{AB}) = P(\boldsymbol{A})\,P(\boldsymbol{B})$$

Section 2.7 The independence of random variables

Definition 2.7.1: Suppose X and Y are two random variables, and $F_X(x)$ and $F_Y(y)$ are their marginal distribution functions, F(x, y) is joint cdf of two-dimensional r.v. (X, Y). If **for any x, y:**

$$P\{X \le x, Y \le y\} = P\{X \le x\}P\{Y \le y\}$$

we say X and Y are independent. Otherwise X and Y are dependent. It can also be marked by following CDF:

$$F(x, y) = F_X(x)F_Y(y)$$

Note: When random variables X and Y are independent, the joint CDF of r.v. (X,Y) is the product of its two marginal CDF.

$$X,Y \text{ are independent} \leftrightarrow F(x, y) = F_X(x)F_Y(y)$$

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Especially for discrete variables:

If X and Y are two discrete random variables, and $P\{X = x_i\} = p_{i\cdot}$ and $P\{Y = Y_j\} = p_{\cdot j}$ are their probability distributions, and $P\{X = x_i, Y = y_j\} = p_{ij}$ is joint probability distribution of (X, Y), then X and Y are independent if and only if

$$P\{X = x_i, Y = y_j\} = P\{X = x_i\}P\{Y = Y_j\}$$

$$(\text{i.e.} \quad p_{ij} = p_{i\cdot}p_{\cdot j})$$

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Example. Let X and Y be independent of each other, and their distribution laws are:

X	0	1
P	$\frac{2}{3}$	$\frac{1}{3}$

Y	1	2	3
P	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Find: joint distribution law of **(X,Y).**

Solution:

X \ Y	1	2	3
0	$\frac{2}{12}$	$\frac{4}{12}$	$\frac{2}{12}$
1	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{1}{12}$

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Example 2.7.1: The distribution law of (X,Y) is

(X,Y)	(1,1)	(1,2)	(1,3)	(2,1)	(2,2)	(2,3)
p_{ij}	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{1}{18}$	$\frac{1}{3}$	α	β

(1) What is the relationship between α and β ?

(2) When X and Y are independent, calculate the values of α and β.

Solution:

(1) $\alpha > 0$ and $\beta > 0$ and $\frac{2}{3} + \alpha + \beta = 1$

(2) X and Y are independent, then:

$$\alpha = \frac{1}{9};\ \beta = \frac{1}{9}$$

$X \backslash Y$	1	2	3	$p_{i\bullet} = P\{X = x_i\}$
1	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{1}{18}$	$\frac{1}{3}$
2	$\frac{1}{3}$	α	β	$\frac{1}{3} + \alpha + \beta$
$p_{\bullet j} = P\{Y = y_j\}$	$\frac{1}{2}$	$\frac{1}{9} + \alpha$	$\frac{1}{18} + \beta$	$\frac{2}{3} + \alpha + \beta$

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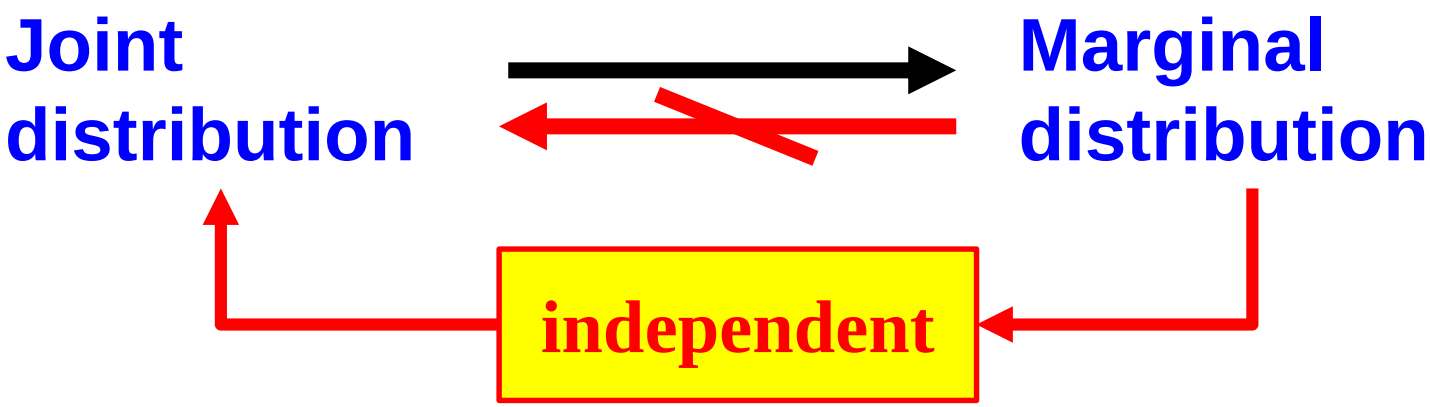
- For discrete random variables, if the joint distribution is known, the corresponding marginal distribution can be obtained, but on the contrary, it is not.
- However, once the independent conditions of X and Y are known, the joint distribution can be obtained directly from the marginal distribution.



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Especially for continuous variables:

If X and Y are two continuous random variables, and $f_X(x)$ and $f_Y(y)$ are their probability density functions and $f(x, y)$ is joint PDF of (X, Y), then X and Y are independent if and only if

$$f(x, y) = f_X(x) f_Y(y)$$

this equality hold almost everywhere in $\mathbb{R}^2$.
Remark: "this equality hold almost everywhere in$\mathbb{R}^2$" means that this equality hold outside a subset of $\mathbb{R}^2$, and this subset's measure or area is 0.

Section 2.7 The independence of r.v.

Example 2.7.2: The PDF of r.v. (X,Y) is

$$f(x,y)=\begin{cases} xe^{-(x+y)}, x>0, y>0 \\ 0\,, \qquad others \end{cases}$$

Are X and Y independent?

Solution:

$$f_X(x)=\int_0^{\infty} xe^{-(x+y)}dy = \boldsymbol{xe^{-x}} \qquad x>0$$

$$f_Y(y)=\int_0^{\infty} xe^{-(x+y)}dx = \boldsymbol{e^{-y}} \qquad y>0$$

$$f_X(x)=\begin{cases} xe^{-x}, & x>0 \\ 0, & others \end{cases} \qquad f_Y(y)=\begin{cases} e^{-y}, & y>0 \\ 0, & others \end{cases}$$

$$f(x,y)=f_X(x)f_Y(y) \text{ independent!}$$

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Example Suppose that random variables (X, Y) obey uniform distribution in rectangular domain: $a \le x \le b\,, c \le y \le d$

Find: (1) joint pdf and marginal pdf of (X,Y);
(2) show that X and Y are independent of each other;
(3) Joint distribution function of (X, Y);
(4) $P(X \le b, Y \le \frac{c+d}{2})$

Solution: **(1)** joint pdf of (X, Y) is:

$$f(x,y) = \begin{cases} \dfrac{1}{(b-a)(d-c)} & a \le x \le b, c \le y \le d \\ 0 & \text{others} \end{cases}$$

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Example

In the region of $a \le x \le b, c \le y \le d$, it has:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)\,dy = \int_c^d \frac{dy}{(b-a)(d-c)} = \frac{1}{b-a}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y)\,dx = \int_a^b \frac{dx}{(b-a)(d-c)} = \frac{1}{d-c}$$

Out of this region, $f_X(x) = 0$, $f_Y(y) = 0$

$$f_X(x) = \begin{cases} \dfrac{1}{b-a} & a \le x \le b \\ 0 & \text{others} \end{cases} \qquad f_Y(y) = \begin{cases} \dfrac{1}{d-c} & c \le y \le d \\ 0 & \text{others} \end{cases}$$

Example

(2) Show that X and Y are independent of each other;

$$f_X(x)\cdot f_Y(y)=\begin{cases}\dfrac{1}{b-a}\cdot\dfrac{1}{d-c} & a\le x\le b,\ c\le y\le d\\ 0 & \text{others}\end{cases}=f(x,y)$$

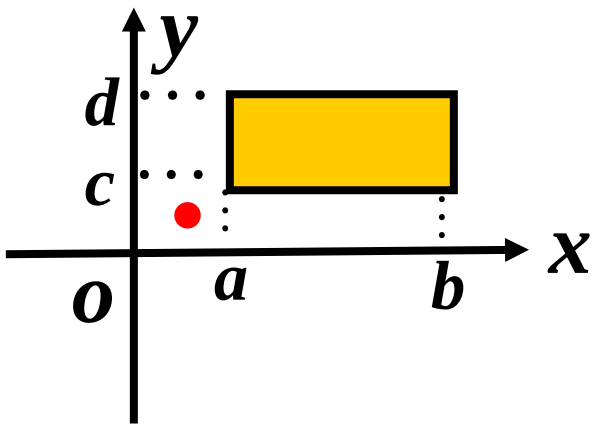
$\Rightarrow$ X and Y are independent of each other.

(3) Joint distribution function of (X, Y);

When $x<a$ or $y<c$

$$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dudv$$

$$=\int_{-\infty}^{x}\int_{-\infty}^{y} o\,dudv=0$$



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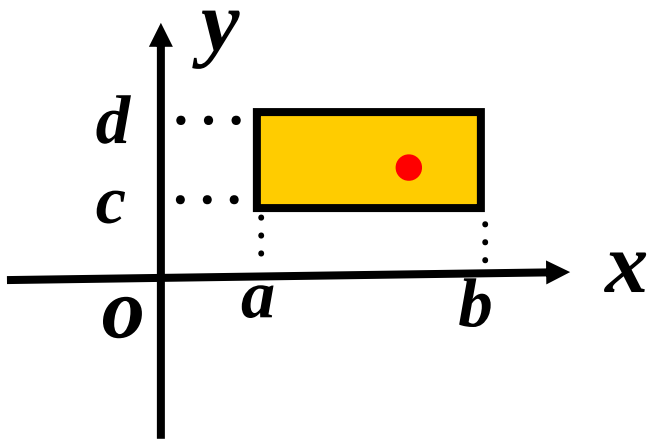
Example

When $a \le x \le b,\ c \le y \le d$

$$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dudv$$

$$=\int_{a}^{x}\int_{c}^{y}\frac{1}{(b-a)(d-c)}\,dudv$$

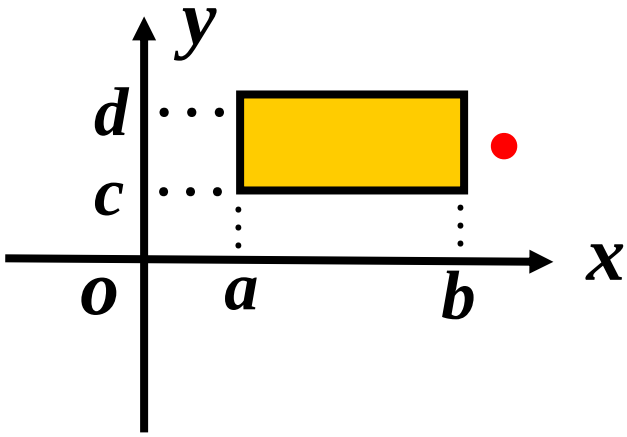
$$=\frac{(x-a)(y-c)}{(b-a)(d-c)}$$



When $x > b,\ c \le y \le d$

$$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dudv$$

$$=\int_{a}^{b}\int_{c}^{y}\frac{1}{(b-a)(d-c)}\,dudv=\frac{(y-c)}{(d-c)}$$



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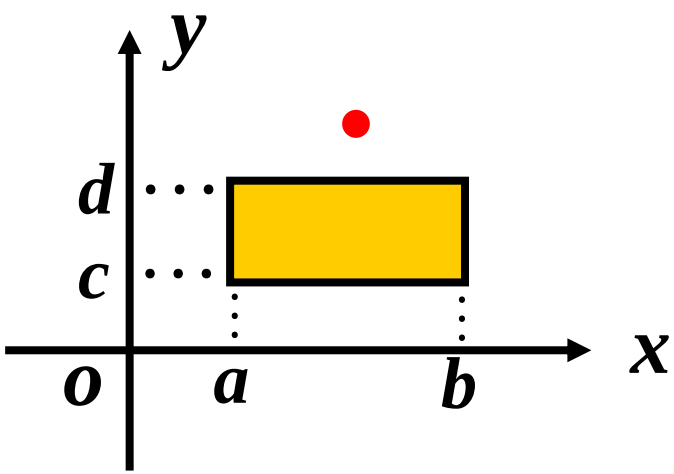
Example

When $y > d$, $a \le x \le b$

$$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dudv$$

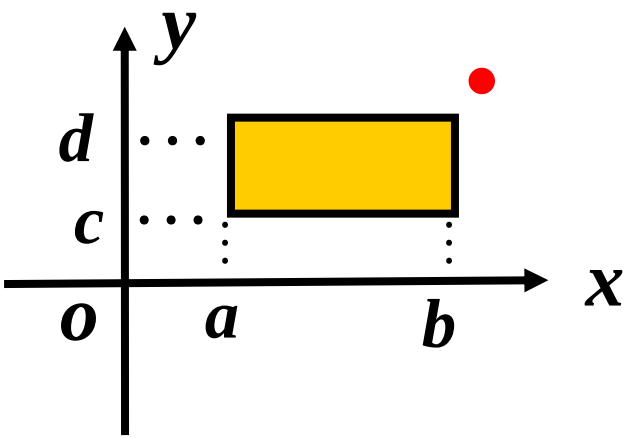
$$=\int_{c}^{d}\int_{a}^{x}\frac{1}{(b-a)(d-c)}\,dudv=\frac{(x-a)}{(b-a)}$$



When $x > b$, $y > d$

$$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dudv$$

$$=\int_{a}^{b}\int_{c}^{d}\frac{1}{(b-a)(d-c)}\,dudv=1$$



Example

So, joint distribution function of (X, Y) is :

$$F(x,y)=\begin{cases} 0 & x<a \text{ or } y<c \\ \dfrac{(x-a)(y-c)}{(b-a)(d-c)} & a\le x\le b, c\le y\le d \\ \dfrac{y-c}{d-c} & x>b, c\le y\le d \\ \dfrac{x-a}{b-a} & y>d, a\le x\le b \\ 1 & x>b \ \& \ y>d \end{cases}$$

Example

(4). $P(X \le b, Y \le \frac{c+d}{2})$

$$P(X \le b, Y \le \frac{c+d}{2}) = F(b, \frac{c+d}{2})$$

$$= \frac{(b-a)(\frac{c+d}{2}-c)}{(b-a)(d-c)} = \frac{1}{2}$$

Section 2.7 The independence of random variables

Example. Find the conditions which makes $\boldsymbol{X}$ and $\boldsymbol{Y}$ independent for bivariate normal distribution $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$.

In fact,

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}[\frac{(x-\mu_1)^2}{\sigma_1^2} -2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}]\}$$

and

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}, f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(y-\mu_2)^2}{2\sigma_2^2}}$$

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In fact,

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\exp\{-\frac{1}{2(1-\rho^2)}[\frac{(x-\mu_1)^2}{\sigma_1^2}$$

$$-2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{\sigma_2^2}]\}$$

and

$$f_X(x)=\frac{1}{\sqrt{2\pi}\sigma_1}\mathrm{e}^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}},\ f_Y(y)=\frac{1}{\sqrt{2\pi}\sigma_2}\mathrm{e}^{-\frac{(y-\mu_2)^2}{2\sigma_2^2}}$$

If $f_X(x)\cdot f_Y(y)=f(x,y)$ Then, $\rho=0$

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In fact,

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\exp\{-\frac{1}{2(1-\rho^2)}[\frac{(x-\mu_1)^2}{\sigma_1^2}$$

$$-2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{\sigma_2^2}]\}$$

and

$$f_X(x)=\frac{1}{\sqrt{2\pi}\sigma_1}\mathrm{e}^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}},\ f_Y(y)=\frac{1}{\sqrt{2\pi}\sigma_2}\mathrm{e}^{-\frac{(y-\mu_2)^2}{2\sigma_2^2}}$$

Conversely, when $\rho = 0$, the above equality is hold obviously, so $f(x, y) = f_X(x)f_Y(y)$.

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In fact,

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\exp\{-\frac{1}{2(1-\rho^2)}[\frac{(x-\mu_1)^2}{\sigma_1^2}$$

$$-2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{\sigma_2^2}]\}$$

and

$$f_X(x)=\frac{1}{\sqrt{2\pi}\sigma_1}\mathrm{e}^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}},\ f_Y(y)=\frac{1}{\sqrt{2\pi}\sigma_2}\mathrm{e}^{-\frac{(y-\mu_2)^2}{2\sigma_2^2}}$$

Finally we get a conclusion: For bivariate normal distribution $(\boldsymbol{X}, \boldsymbol{Y}) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, $\boldsymbol{X}$ and $\boldsymbol{Y}$ are independent if and **only if $\rho = 0$.**

Section 2.7 The independence of random variables

The concept that *n*-dimensional random variables are independent of each other

Definition 1.

If for any $x_1, x_2, \cdots, x_n$, it has:

$$F(x_1, x_2, \cdots, x_n) = F_{X_1}(x_1) \cdot F_{X_2}(x_2) \cdots F_{X_n}(x_n)$$

Then, $X_1, X_2, \cdots, X_n$ are independent of each other .

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The concept that *n*-dimensional random variables are independent of each other

Definition 2.

If for any $x_1, x_2, \cdots, x_m; y_1, y_2, \cdots, y_n$, it has：

$$F(x_1, x_2, \cdots, x_m; y_1, y_2, \cdots, y_n)$$
$$= F_1(x_1, x_2, \cdots, x_m) \cdot F_2(y_1, y_2, \cdots, y_n)$$

where F_1, F_2, F are the distribution functions of

$(X_1, X_2, \cdots, X_m)$, $(Y_1, Y_2, \cdots, Y_n)$ and $(X_1, X_2, \cdots, X_m; Y_1, Y_2, \cdots, Y_n)$

then, $(X_1, X_2, \cdots, X_m)$ and $(Y_1, Y_2, \cdots, Y_n)$ are independent of each other.

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Theorem 1.

The pdf of continuous r.v. (X_1, …,X_n) can be expressed as the product of n functions g_1, …, g_n . And here g_i depends only on X_i, that is :

$$f(x_1, x_2, \cdots, x_n) = g_1(x_1) \cdot g_2(x_2) \cdots g_n(x_n)$$

Then $X_1, X_2, \cdots, X_n$ are independent of each other, and the marginal pdf of X_i($f_i(x_i)$) is only different from $g_i(x_i)$ by a constant factor.

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Theorem 2.

If $X_1, \ldots, X_n$ are independent of each other , and

$Y_1 = g_1(X_1, \ldots, X_m)$, $Y_2 = g_2(X_{m+1}, \ldots, X_n)$,

Then Y_1 and Y_2 are independent of each other.

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Theorem 3.

Suppose $(X_1, X_2, \cdots, X_m)$ and $(Y_1, Y_2, \cdots, Y_n)$ are independent of each other, then $X_i\ (i = 1, 2, \cdots, m)$ and $Y_j\ (j = 1, 2, \cdots, n)$ are independent of each other .

And if h and g are continuous function， then $h(X_1, X_2, \cdots, X_m)$ and $g(Y_1, Y_2, \cdots, Y_n)$ are independent of each other.